

Temporal variation in a savanna bird assemblage: what changes over 5 years?

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Abstract. Tropical savanna environments are characterised by annual and decadal patterns of resource change, which can affect the patterning of mobile fauna such as birds. In this study, we sampled 60 sites in northern Queensland, four times from 2004 to 2008. We investigated how the bird richness and abundance, and species turnover changed over the sample years and how this differed with vegetation structure. The mean abundance per site was highest in 2005 (92.1 ± 12.0 individuals ha^{-1}) and lowest in 2008 (46.6 ± 3.3), whereas species richness per site was highest in 2004 (19.6 ± 0.9 species ha^{-1}) and lowest again in 2008 (14.7 ± 0.8). Nine species were most abundant in 2004 coincident with extremely high rainfall preceding the survey in that year. Species turnover increased across all sites from 2004 to 2008 and the abundance of 13 species was best accounted for by differences in vegetation structure. Our study demonstrates that local bird communities in the semi-arid fringe of savannas can change rapidly, and mostly where vegetation is modified. This suggests that increased land-use and climate change in northern Australia could have significant effects on the avifauna over fairly short periods.

Additional keywords: climate, conservation, habitat modification, migratory, nomadic, sedentary, species turnover, tropical.

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Introduction

There has been a shift in our understanding of how avian species assemble across the landscape, from an expectation that they are largely saturated, equilibrium populations (Recher 1969) to acceptance that assemblage structure changes over time and is not at equilibrium (Rittenhouse *et al.* 2010). Most intact landscapes and the avifauna within them form continuous gradients of individuals and resources (Price *et al.* 2009). In some biomes, such as tropical savanna regions, large variations in resource availability are a natural component of the annual or decadal cycles in climate (Holmgren *et al.* 2001). As such, many avian populations can change in response to long-term climatic cycles (Albright *et al.* 2010), although anthropogenic effects, such as increasing urbanisation (Radeloff *et al.* 2010) and long-term changes in land management (Schulte *et al.* 2005) are also significant determinants of avifaunal assemblages.

Although there are several predictions of how bird communities change annually (Woinarski *et al.* 2000) and over decades (Reside *et al.* 2010) in Australian tropical savannas, there are very few examples of studies examining intermediate time-frames. Further, many correlative studies of the relative effect of climate or vegetation change on the structure of avian assemblages are based on annual or seasonal data (Maron *et al.* 2005; Tassicker *et al.* 2006; Hannah *et al.* 2007; Pavey and Nano 2009). Recent examination of historical patterns of change in the distribution of

Australian birds has shown that there are rapid rates of change in climatic niche space (average $>1 \text{ km year}^{-1}$), and this change in niche space occurs at a faster annual rate than recently recorded annual changes in temperature or precipitation over the past 60 years (J. VanDerWal, unpubl. data). Land management, such as use of fire and grazing of introduced stock, can also have a compounding effect on abundance of birds and avian community composition owing to changes in woody and ground-cover vegetation and, therefore, habitat resources (Kutt and Fisher 2011).

Thus, understanding change in bird assemblage pattern at intermediate time-frames (>1 year and <10 years) and scales (property, region or ecosystem type), and also where landscapes are altered by human land management, is important for setting conservation actions in an era of rapid global change. In this study we examine a discrete set of sites for signs of change in avian assemblages, within a small region of the tropical savannas that were sampled systematically over a 5-year period. The sites were located in a single vegetation type, but represent a range of intact and modified (i.e. cleared, thinned) vegetation. These data allow us to test three complementary questions regarding the pattern and change in avifauna in tropical savannas: (1) that changes in climatic pattern may drive changes in the species richness and abundance of birds within assemblages over an intermediate time period; (2) the magnitude of changes to the avifauna will

be greater in modified vegetation; and (3) the largest changes in turnover and composition of species will be in guilds that are attuned to responding to resource variation (i.e. nomadic species) compared with sedentary species.

Materials and methods

Study area

The study area is in the Desert Uplands bioregion, at the xeric southern reaches of Queensland's tropical savanna. The vegetation of the region is predominantly *Eucalyptus* and *Acacia* woodlands, open grasslands and ephemeral lakes (Price *et al.* 2010). All sites sampled in this study were located in extensive ironbark (*Eucalyptus whitei*, *E. melanophloia*) woodlands in the north-western section of the Desert Uplands bioregion (Fig. 1). The region's annual rainfall is influenced by monsoonal weather patterns, with most of the annual average rainfall (350 mm) falling between December and March.

Sampling design

Sixty sites, each 1 ha, were sampled for birds over a period of 2 weeks in June 2004, May 2005, August 2006 and July 2008, all sampling occurring in the middle of the dry season. For each 1-ha

site, eight 5-min wandering diurnal bird counts were conducted over 4 days. At least two bird counts were made per day at each 1-ha site: one count in the morning between dawn and 3 hours after dawn, and the other a minimum of 3 hours after this time and before dusk. This survey method has been used extensively in northern Australia's tropical savannas (Kutt and Woinarski 2007). Data used in the analyses were the total summed abundance of all eight counts for each site for each year. All sites were a minimum of 500 m from watering points to standardise the effect of grazing pressure across sites, and as far as practical from fencelines and roads. Sites were at least 1 km apart to avoid spatial dependence, and all sites were within a discrete unit of a mapped vegetation polygon (i.e. away from edges of the polygon). Within the 60 sites, three vegetation structural types were represented and sampled: cleared vegetation ($n = 10$ sites), mechanically thinned vegetation ($n = 10$), and unmodified vegetation ($n = 40$). These sites were spread over four properties: 10 cleared, 10 intact on one property; 20 intact on the second; and 10 thinned and 10 intact on the third and fourth properties. Detailed information on the site stratification can be found in Tassicker *et al.* (2006). The vegetation structural types were not all fully interspersed (i.e. all three vegetation structural types were not on each property sampled), as the spatial distribution of thinning and clearing within the single vegetation type sampled was limited. However, we believe that the focus on a single vegetation type, coupled with the large number of sites, themselves widely dispersed throughout the properties (i.e. 1 km apart and spread throughout an area of 10–15-km radius on each property), captures and accounts for natural variation in species distribution. Data collection at a landscape scale that uses the natural mosaic of vegetation as natural experimental plots provides information on patterns and responses of wildlife at a scale relevant to the conservation manager (Lindenmayer *et al.* 2008).

Analysis

The variation in bird species composition among years and within vegetation structural type (and the interaction) was tested by (PERMANOVA, permutational multivariate analysis of variance; Anderson *et al.* 2008) using Bray–Curtis dissimilarity matrices after a square-root transformation of the data. The variation in total abundance of birds, species richness and abundance of individual species among the four survey years was tested using a one-way analysis of variance (ANOVA). Only species that were recorded from more than five sites in each survey year were examined.

As the four survey periods were not in consecutive years we calculated the minimum species turnover rate (T) for each site using the formula:

$$T = (E_{spp} + C_{spp}) / S_{tot} \times 100$$

where E_{spp} is the number of local extinctions (number of species not recorded at the site in year 2 that had been recorded during year 1), C_{spp} is number of local colonisations (number of species recorded at the site during survey period 2 that had not been recorded in survey period 1) and S_{tot} the total number of species observed at the site during both survey periods (Maron *et al.* 2005). Mean minimum turnover per site was calculated for all species combined, for adjacent survey years, and for movement

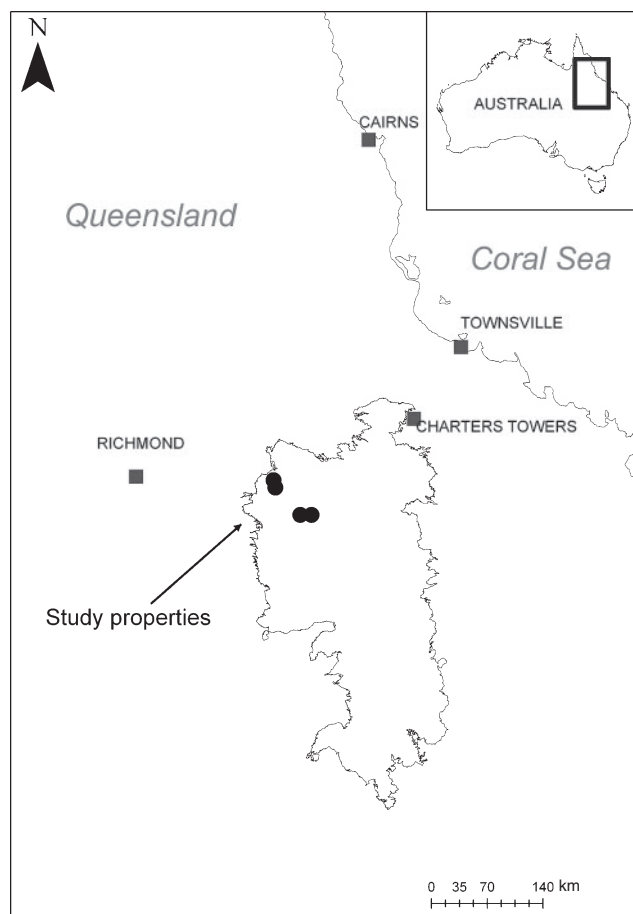


Fig. 1. The Desert Uplands bioregion of north-eastern Queensland, Australia, with the properties on which survey sites located indicated by large solid circles.

categories of species (nomadic, sedentary, migratory) defined following Reside *et al.* (2010). Mean minimum turnover was also calculated for vegetation structure treatment subsets. We then undertook paired *t*-tests for comparisons between years (as we were only comparing changes from year to year), and between movement categories for all sites and for subsets of the sites representing structural treatments.

Lastly we used a repeated-measures ANOVA to test the variation in total abundance, species richness and species abundance between years, within treatments (and the interaction). All ANOVA were conducted with the R v2.9.0 software program (R Development Core Team 2011 <http://www.r-project.org/>). Patterns of bird abundance in tropical savannas are strongly associated with both vegetation pattern and rainfall (Pavey and Nano 2009; Perry *et al.* 2011). To provide context for the annual changes in the patterns of abundance recorded, we examined annual rainfall totals before each survey (http://www.bom.gov.au/climate/averages/tables/ca_qld_names.shtml, accessed 1 December 2010). The annual precipitation in the study area ranged from 347 to 717 mm over the 4 years of study, and the annual average for the 4 years is 545 mm. Survey period 2004 was 198 mm below the average rainfall, 2005 was 70 mm below average, 2006 on average and 2008 169 mm above the yearly average. The combined rainfall totals for the 3 consecutive months leading up to each field trip was: 220 mm (2004), 287 mm (2005), 83 mm (2006) and 14 mm (2008). As we use year as one of the repeated-measures terms and that rainfall preceding each survey varied greatly in each year, rainfall and year are in effect interchangeable terms: any significant effect of year on abundance also represents a significant effect of rainfall change. As such we do not test a rainfall effect specifically, as this would be redundant.

Results

A total of 110 species, of 38 families, was recorded across all the surveys, with >15 000 individual records. Total abundance ranged from a maximum of 5460 birds in 2005 to a minimum of 2799 in 2008. Overall species richness ranged from 77 species in 2008 to 82 in 2004 and 2005. The most abundant species recorded were (in decreasing order of abundance): Weebill (*Smicrornis brevirostris*), Budgerigar (*Melopsittacus undulatus*), Zebra Finch (*Taeniopygia guttata*), Yellow-throated Miner (*Manorina flavigula*), Striated Pardalote (*Pardalotus striatus*), Variegated Fairy-wren (*Malurus lamberti*), Double-barred Finch (*Taeniopygia bichenovii*), Yellow-rumped Thornbill (*Acanthiza chrysorrhoa*), Rufous Whistler (*Pachycephala rufiventris*) and Singing Honeyeater (*Lichenostomus virescens*) (Table 1).

The results of the PERMANOVA indicated significant variation in avian community composition across years (pseudo-*F* = 6.6, *P* = 0.001), between vegetation structural types (pseudo-*F* = 6.5, *P* = 0.001), and the interaction (pseudo-*F* = 2.1, *P* = 0.001).

The mean ($\pm$ s.e.) abundance per site was highest in 2005 (92.1 ± 12.0 birds per site), and lowest in 2008 (46.6 ± 3.3), whereas species richness per site was highest in 2004 (19.6 ± 0.9 species per site) and lowest, again, in 2008 (14.7 ± 0.8) (Table 1). Nine species were more abundant in 2004 than any other year (e.g. Grey-fronted Honeyeater (*Lichenostomus plumulus*), Pied Butcherbird (*Cracticus quoyi*), Singing Honeyeater); in subsequent years only five species (in 2005, e.g. Budgerigar, Double-barred Finch), three species (in 2006, e.g. Inland Thornbill (*Acanthiza apicalis*), White-winged Triller (*Lalage sueurii*)) and two species (2008, e.g. Weebill, Grey Butcherbird (*Cracticus torquatus*)) recorded a significantly higher mean abundance (Table 1).

Table 1. Mean ($\pm$ standard error) total abundance per site, species richness and abundance of individual species for each year of the survey. Only species that showed significant change between years (identified by one-way ANOVA, *P* < 0.05) are listed. The highest totals are in underlined

Species	2004	2005	2006	2008	<i>P</i>
Mean abundance (mean number of birds per site)	72.2 $\pm$ 4.6	<u>92.1 $\pm$ 12.0</u>	56.3 $\pm$ 3.3	46.6 $\pm$ 3.3	<0.001
Mean species richness (mean number of species per site)	<u>19.6 $\pm$ 0.9</u>	17.3 $\pm$ 0.7	17.3 $\pm$ 0.8	14.7 $\pm$ 0.8	<0.001
Peaceful Dove (<i>Geopelia striata</i>)	<u>0.5 $\pm$ 0.1</u>	0.1 $\pm$ 0.1	0.4 $\pm$ 0.1	0.2 $\pm$ 0.1	0.020
Brown Falcon (<i>Falco berigora</i>)	0.3 $\pm$ 0.1	<u>0.4 $\pm$ 0.1</u>	0.1 $\pm$ 0.0	0.3 $\pm$ 0.1	0.024
Pale-headed Rosella (<i>Platycercus adscitus</i>)	<u>1.1 $\pm$ 0.3</u>	0.4 $\pm$ 0.1	0.8 $\pm$ 0.2	0.4 $\pm$ 0.1	0.020
Budgerigar (<i>Melopsittacus undulatus</i>)	0.6 $\pm$ 0.3	<u>15.2 $\pm$ 5.8</u>	0.8 $\pm$ 0.5	0.2 $\pm$ 0.2	0.003
Variegated Fairy-wren (<i>Malurus lamberti</i>)	<u>4.0 $\pm$ 0.7</u>	2.2 $\pm$ 0.5	2.6 $\pm$ 0.5	1.7 $\pm$ 0.4	0.026
Weebill (<i>Smicrornis brevirostris</i>)	10.2 $\pm$ 1.3	8.8 $\pm$ 0.7	6.4 $\pm$ 0.8	<u>11.5 $\pm$ 1.0</u>	0.003
Inland Thornbill (<i>Acanthiza apicalis</i>)	0.1 $\pm$ 0.1	0.4 $\pm$ 0.2	<u>0.8 $\pm$ 0.3</u>	0.2 $\pm$ 0.1	0.007
Singing Honeyeater (<i>Lichenostomus virescens</i>)	<u>2.6 $\pm$ 0.3</u>	1.6 $\pm$ 0.3	1.7 $\pm$ 0.2	1.8 $\pm$ 0.2	0.030
Grey-fronted Honeyeater (<i>Lichenostomus plumulus</i>)	<u>4.0 $\pm$ 1.4</u>	1.5 $\pm$ 0.6	1.0 $\pm$ 0.3	1.0 $\pm$ 0.3	0.016
Spiny-cheeked Honeyeater (<i>Acanthagenys rufogularis</i>)	<u>1.7 $\pm$ 0.4</u>	0.4 $\pm$ 0.1	0.6 $\pm$ 0.2	0.6 $\pm$ 0.1	<0.001
Noisy Friarbird (<i>Philemon corniculatus</i>)	<u>0.4 $\pm$ 0.1</u>	0.1 $\pm$ 0.1	0.0	0.2 $\pm$ 0.1	0.010
Striped Honeyeater (<i>Plectorhyncha lanceolata</i>)	<u>1.0 $\pm$ 0.2</u>	0.5 $\pm$ 0.1	0.3 $\pm$ 0.1	0.4 $\pm$ 0.1	0.002
Varied Sittella (<i>Daphoenositta chrysoptera</i>)	0.7 $\pm$ 0.2	<u>1.2 $\pm$ 0.3</u>	1.0 $\pm$ 0.3	0.3 $\pm$ 0.1	0.049
Black-faced Cuckoo-shrike (<i>Coracina novaehollandiae</i>)	0.7 $\pm$ 0.1	1.1 $\pm$ 0.3	<u>1.2 $\pm$ 0.25</u>	0.2 $\pm$ 0.1	0.007
White-winged Triller (<i>Lalage sueurii</i>)	0.2 $\pm$ 0.1	0.3 $\pm$ 0.2	<u>0.8 $\pm$ 0.2</u>	0.0	0.001
Crested Bellbird (<i>Oreoica gutturalis</i>)	<u>1.7 $\pm$ 0.3</u>	1.1 $\pm$ 0.2	1.4 $\pm$ 0.2	0.6 $\pm$ 0.1	0.003
Grey Butcherbird (<i>Cracticus torquatus</i>)	1.2 $\pm$ 0.2	0.8 $\pm$ 0.2	0.8 $\pm$ 0.1	<u>1.9 $\pm$ 0.2</u>	<0.001
Pied Butcherbird (<i>Cracticus nigrogularis</i>)	<u>1.3 $\pm$ 0.2</u>	0.7 $\pm$ 0.2	0.7 $\pm$ 0.1	0.5 $\pm$ 0.1	0.002
Zebra Finch (<i>Taeniopygia guttata</i>)	1.9 $\pm$ 0.1	<u>11.7 $\pm$ 4.9</u>	1.8 $\pm$ 0.7	0.2 $\pm$ 0.1	0.005
Double-barred Finch (<i>Taeniopygia bichenovii</i>)	2.3 $\pm$ 0.6	<u>4.4 $\pm$ 1.0</u>	1.9 $\pm$ 0.5	1.8 $\pm$ 0.6	0.024

Species turnover generally increased across all sites from 2004 to 2008, as did the turnover within the three movement categories (Table 2). The pattern was more variable when sites were analysed by vegetation structural category: in cleared sites, species turnover increased in each movement category, but most dramatically in nomadic species; in thinned sites, the turnover decreased gradually for all species over time, and most dramatically for sedentary and migratory species; and for unmodified sites, the changes over time match the changes observed for all sites, with gradual increases from 2004 to 2008, and mostly for migratory species (Table 2).

Variation in species turnover among years was fairly weak, even within the different structural subsets. The strongest variation in turnover was between sedentary versus nomadic species in all sites (paired $t = -10.15$, $P < 0.01$); migratory versus nomadic species in all sites ($t = -4.50$, $P < 0.01$) and unmodified sites ($t = -5.26$, $P < 0.01$); and sedentary versus nomadic species in thinned ($t = -7.68$, $P < 0.01$) and unmodified sites ($t = -8.41$, $P < 0.01$) (Table 3).

Finally, the abundance of a large number of species varied over the years of sampling within the different structural types, across the years of survey and with the interaction of these two terms (Table 4). Abundance varied significantly on an annual basis, although there was an interaction with vegetation structure. The significant variation in species richness was predicted by year of survey, vegetation structure and the interaction term. The abundance of 13 species varied significantly with vegetation structure, the abundance of 11 species changed with year of sampling and abundance of seven were related to the interaction term (Table 4). Most of the variation in species abundance was accounted for by vegetation structure and the year of study (i.e. rainfall), although abundance of Black-faced Cuckoo-shrikes (*Coracina novaehollandiae*) was significantly related to year of survey, whereas abundance of Double-barred Finches, Noisy Friarbirds (*Philemon corniculatus*), Pale-headed Rosellas (*Platycercus adscitus*), Peaceful Doves (*Geopelia striata*), Spiny-cheeked Honeyeaters (*Acanthagenys rufogularis*) and Striped Honeyeaters (*Plectorhyncha lanceolata*) was most strongly related to differences in vegetation structure. The abundance of Pied Butcherbirds and

Singing Honeyeaters varied across years, although interacting with the vegetation structure.

Discussion

The bird assemblage that we sampled four times over a period of 5 years changed in composition over time and within the different vegetation structures sampled. There were clear shifts in species abundance (mainly declines) over the survey period and changes in species and functional groups recorded on an annual basis, but this varied according to vegetation structure. This suggests that although bird communities of tropical savannas were considered stable in composition and structure over time and space (Woinarski *et al.* 2005), yearly and decadal changes do occur in response to factors such as changing weather patterns (Reside *et al.* 2010) that can then be amplified where there was mechanical alternation of woodland structure (Kutt and Martin 2010). This might result in a gradual depletion of species over time, where the landscape becomes more heavily modified via actions such as land clearing and resulting fragmentation of habitat (Hannah *et al.* 2007). This contrasts with results from intact savanna vegetation that found little change in the species composition and abundance over 9 years (Perry *et al.* 2011) and severe declines of birds in heavily modified landscapes resurveyed over a period of 80 years in central Queensland (Woinarski and Catterall 2004).

Our study shows that the rates of species turnover in cleared sites (more often than not the mechanically thinned sites) were generally always higher than in unmodified sites, and that turnover of nomadic species increased markedly over the survey period. For thinned sites, migratory species declined over the years compared with the other vegetation structural types (cleared, unmodified). These results suggest that where vegetation structure has been physically altered from its natural state, there was a change in the stability of the bird community. This is commensurate with greater species turnover in highly fragmented woodlands observed in south-eastern Australia (Maron *et al.* 2005), and reduced species richness and abundance where the canopy and mid-story has been mechanically thinned (Kutt 1996), or partially cleared (Hannah *et al.* 2007). At a broader scale

Table 2. The mean minimum rate of species turnover per site ($\pm$ standard error) for each year of the survey
Data are presented for all species and the three movement categories identified (sedentary, nomadic and migratory) for each vegetation structural types (clear, thinned and unmodified)

Category	Year comparison	<i>n</i>	All sites (<i>n</i> = 60)	Cleared (<i>n</i> = 10)	Thinned (<i>n</i> = 10)	Unmodified (<i>n</i> = 40)
All species	2004 v. 2005	60	66 $\pm$ 1	71 $\pm$ 2	70 $\pm$ 3	64 $\pm$ 1
	2005 v. 2006	60	72 $\pm$ 1	80 $\pm$ 3	70 $\pm$ 3	71 $\pm$ 2
	2006 v. 2008	60	71 $\pm$ 2	84 $\pm$ 3	65 $\pm$ 3	71 $\pm$ 2
Sedentary	2004 v. 2005	58	61 $\pm$ 2	66 $\pm$ 6	62 $\pm$ 4	60 $\pm$ 2
	2005 v. 2006	60	67 $\pm$ 2	78 $\pm$ 4	61 $\pm$ 4	67 $\pm$ 3
	2006 v. 2008	60	67 $\pm$ 2	78 $\pm$ 4	61 $\pm$ 4	67 $\pm$ 3
Nomadic	2004 v. 2005	59	77 $\pm$ 2	70 $\pm$ 4	85 $\pm$ 5	78 $\pm$ 2
	2005 v. 2006	60	80 $\pm$ 2	80 $\pm$ 4	85 $\pm$ 4	79 $\pm$ 3
	2006 v. 2008	60	84 $\pm$ 2	90 $\pm$ 4	83 $\pm$ 4	83 $\pm$ 2
Migratory	2004 v. 2005	60	64 $\pm$ 4	91 $\pm$ 6	77 $\pm$ 11	56 $\pm$ 4
	2005 v. 2006	60	71 $\pm$ 4	93 $\pm$ 7	68 $\pm$ 10	66 $\pm$ 4
	2006 v. 2008	60	72 $\pm$ 4	90 $\pm$ 10	58 $\pm$ 12	71 $\pm$ 5

Table 3. The results of paired *t*-test comparisons of mean minimum species turnover between years and between bird movement categories
Significance levels are: ns, not significant; *, $P < 0.05$; **, $P < 0.001$

Comparison	All		Clear		Thinned		Unmodified	
	d.f.	<i>t</i>	d.f.	<i>t</i>	d.f.	<i>t</i>	d.f.	<i>t</i>
2004–05 v. 2005–06	118	−3.07*	18	−2.26*	18	−0.16 ^{ns}	78	−2.77*
2005–06 v. 2006–08	118	0.13 ^{ns}	18	−0.72 ^{ns}	18	1.10 ^{ns}	78	−0.02 ^{ns}
Sedentary v. nomadic	180	−10.15**	58	−1.52 ^{ns}	58	−7.68**	238	−8.41**
Sedentary v. migratory	355	−1.82 ^{ns}	57	−3.32*	57	−1.16 ^{ns}	237	−0.26 ^{ns}
Migratory v. nomadic	355	−4.50**	57	2.20*	57	−2.49*	237	−5.26**

Table 4. The variation in bird abundance between vegetation structural type, year and the interaction term

Significant variation is identified by repeated-measures ANOVA and only groups or species with significant results are presented; significant probabilities are marked in bold. Scientific names of species are given in Table 1

Group (species)	Structure			Year			Year × Structure		
	d.f.	<i>F</i>	<i>P</i>	d.f.	<i>F</i>	<i>P</i>	d.f.	<i>F</i>	<i>P</i>
Mean abundance per site	2	0.2	0.812	3	11.5	<0.001	6	4.7	<0.001
Mean species richness per site	2	757.8	<0.001	3	3.8	0.011	6	3.1	0.006
Peaceful Dove	2	3.4	0.040	3	1.9	0.124	6	1.7	0.112
Brown Falcon	2	2.5	<0.001	3	4.8	0.006	6	3.6	0.002
Pale-headed Rosella	2	5.1	0.008	3	1.6	0.186	6	0.4	0.835
Budgerigar	2	3.3	0.042	3	11.4	<0.001	6	2.7	0.015
Weebill	2	8.3	<0.0001	3	8.6	<0.001	6	3.5	0.002
Inland Thornbill	2	5.4	0.007	3	0.7	0.007	6	1.7	0.332
Singing Honeyeater	2	0.2	0.754	3	3.2	0.022	6	2.7	0.014
Spiny-cheeked Honeyeater	2	7.5	0.001	3	1.4	0.244	6	1.6	0.128
Noisy Friarbird	2	6.2	0.004	3	0.6	0.602	6	0.9	0.474
Striped Honeyeater	2	8.3	<0.001	3	1.3	0.269	6	1.3	0.225
Black-faced Cuckoo-shrike	2	2.4	0.485	3	10.2	0.011	6	1.5	0.152
White-winged Triller	2	3.5	0.036	3	8.7	<0.001	6	2.4	0.025
Crested Bellbird	2	8.5	<0.001	3	4.7	0.029	6	1.7	0.114
Grey Butcherbird	2	12.1	<0.001	3	4.7	0.003	6	0.6	0.698
Pied Butcherbird	2	0.5	0.595	3	2.8	0.040	6	2.3	0.032
Zebra Finch	2	10.9	<0.001	3	11.0	<0.001	6	6.9	<0.001
Double-barred Finch	2	5.7	0.005	3	0.5	0.678	6	1.9	0.069

where the landscape is largely unmodified, these species changes are probably of lesser importance than in highly cleared and relict environments, where the reduction or loss of species may result in the inability to recolonise and lead to local extinctions (Mac Nally *et al.* 2009) or a shift in the avian assemblage to one dominated by fewer and disturbance-tolerant species, for example, hyper-aggressive birds that control the extant bird assemblage (Maron 2007).

In our study, although the average rainfall was highest in 2006 and 2008, the abundance and species richness in those years was the lowest, the larger totals being in 2004 and 2005 when the preceding 3-month rainfall was extremely high. In highly seasonal savanna environments, recent preceding rainfall can have a significant effect on the spatial and temporal variation in abundance of local birds (Price *et al.* 2009) and short-term weather patterns rather than climatic averages have been clearly shown to strongly influence bird distribution in Australian tropical savannas (Reside *et al.* 2010). The turnover in nomadic species increased over the years of the study in the cleared sites, as rainfall in the 3 months preceding a survey decreased. Granivorous nomadic species, such as the Budgerigar and Zebra Finch, feed in open grasslands and cleared areas, flocking in

areas of recent rainfall where annual seed-flushes occur (Wyndham 1982). In contrast, turnover of migratory species declined in thinned sites but increased in unmodified sites over the years of the survey, suggesting that species such as Black-faced Cuckoo-shrike, Grey-fronted Honeyeater and Spiny-cheeked Honeyeater might use the structurally less-diverse sites when resource conditions are optimal (i.e. recent rainfall), but retreat to more typical and more diverse woodlands in other years. Seasonal shifts in habitat driven by changing availability of resources are common phenomena in tropical and temperate environments (Franklin 1999; Coops *et al.* 2009).

Although the annual rainfall for the 2004 sample was the lowest of all four surveys, a large number of birds were recorded at their highest abundance in 2004, which is likely to be a function of the exceptional rainfall just before the survey period. This correlation between availability of resources and pulses in bird abundance is a typical pattern for highly seasonal tropical savannas (Price *et al.* 2009). Honeyeaters and granivorous species were most prominent, and these species feed on flushes of nectar and seed that occur after good rainfall (Franklin and Noske 1999; Kutt and Woinarski 2007). In 2005, also a period of low rainfall but high rainfall preceding the survey, there was a large influx of

Budgerigars and Zebra Finches, two small but nomadic granivores that undertake large-scale movement when rainfall and food conditions vary (Griffioen and Clarke 2002). However, the interacting effect of vegetation structure on species richness and abundance and, more particularly, for many resource-tracking, migrating or nomadic species (Pavey and Nano 2009) suggest that the entire landscape mosaic is an equally important component of species distribution (Price *et al.* 2009). In some cases where there is less habitat modification (i.e. thinning), the change in vegetation structure (i.e. lower density, more ground cover) will advantage some species with a preference for more open woodlands (Kutt and Martin 2010). However, overall the continued disruption or degradation of the intact woodlands by intensification of land-use, and the creation of a more-fragmented mosaic, will ultimately result in longer term changes and declines in the avifauna (Ford 2011).

Conclusion

Our study shows that bird assemblages in tropical savanna environments can change in composition and abundance over a 5-year period of sampling, despite a general view that savanna avifauna has stable community structure in space and time (Woinarski *et al.* 2005). Most importantly, change in species composition and species turnover was greatest in the most modified vegetation, and this was largely driven by species that were nomadic or regional migrants. This hints at an ultimate mechanism for species declines and assemblage change with changes to land-use and habitat (Kutt and Fisher 2011), and this will be increasingly significant as short-term weather patterns (a key determinant of patterning of savanna bird communities) alter with global climate change (Reside *et al.* 2010). There is an increasing need to track the changes of species over time in response to environmental or management action (Lindenmayer and Likens 2010; Morrison *et al.* 2010). Australia has a poor record of collecting wildlife data at the landscape level when it is not directly associated with an economic outcome (i.e. pastoral monitoring), and although there have been compelling historical resurveys (Woinarski and Catterall 2004) and other novel methods of demonstrating marked changes in vertebrate fauna over time (Szabo *et al.* 2011), on the whole the consistent monitoring record is poor (Likens and Lindenmayer 2011).

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